

Multiple Choice (10 marks)

1. A sound wave with frequency f travels through air at speed v . With what speed will a sound wave with frequency $4f$ travel through the air?
 - (a) v^2
 - (b) $v/4$
 - (c) $v/2$
 - (d) $4v$
 - (e) $4v^2$
2. A 60-ohm electric lamp is left connected to a 240-volt line for 3.00 min. How much energy is taken from the line?
 - (a) 1.2×10^4 Joules
 - (b) 3.6×10^4 Joules
 - (c) 2.2×10^4 Joules
 - (d) 1.72×10^4 Joules
 - (e) 1.9×10^4 Joules
3. As more lamps are connected in parallel in a circuit, the current in the power source
 - (a) remains the same
 - (b) Not enough information to say
 - (c) is not affected
 - (d) triples
 - (e) decreases
4. Two spheres of net charge $+5e$ and $-6e$ briefly come into contact. Afterward, which of the following is a possible combination of net charges for the two spheres?
 - (a) Both of the above
 - (b) $+3e$ and $-4e$
 - (c) $+5e$ and $-6e$
 - (d) $+2.5e$ and $-3.5e$
 - (e) $+4e$ and $-5e$
5. During the time an aircraft produces a sonic boom the aircraft is
 - (a) Breaking the speed of light

- (b) Flying slower than sound
- (c) pulling out of a subsonic dive
- (d) flying faster than sound
- (e) Flying at constant speed

True / False (10 marks)

Answer True or False

6. True or False: The color of light most intense in the solar radiation curve is green, as it corresponds to peak photosynthetic activity.
7. True or False: An electric motor with 75% efficiency can lift a 6 kg mass at a constant speed of 0.3 m/s using 120 V AC at 1.0 Amps.
8. True or False: The act of eating does not affect the speed of the big fish, as its direction and energy remain unchanged.
9. True or False: The resultant force on a body weighing 48 lb with an acceleration of 6 ft/sec² is 24 lb.
10. True or False: The minimum escape speed from Earth at an altitude of 200 km is approximately 6.42 km/s.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the similarities in the emission spectra of the doubly ionized lithium atom (Li⁺⁺) and hydrogen. Explain the factors that lead to the observed wavelength shift and its implications.
12. Analyze the relationship between voltage and current in a resistor, using Ohm's Law, to determine the current when the voltage is increased from 120 volts to 180 volts. Explain your reasoning.

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